# A LOG-SINE ANALOGUE OF EULER'S CONSTANT: THE DECAY RATE OF A SINE PRODUCT AT PRIMES

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ABSTRACT. For prime $p$ we study $P(p) = \prod_{k=2}^{p-1} \sin(\pi p/k)$. Its logarithmic size is $\log|P(p)| = -C_{\mathrm{g}}p + O(p^{1/3}\log^2 p)$, where $C_{\mathrm{g}} = 0.74379486183940054420\ldots$. The rate per factor in the limit is $e^{-C_{\mathrm{g}}} = 0.47530676524152038801\ldots$. With the denominators restricted to $k \le \sqrt{p}$, the logarithmic size is $-(\log 2)\sqrt{p} + O(p^{1/3}\log^2 p)$. The means $1-\gamma$ and $C_{\mathrm{g}}$ are obtained from two observables of the same fractional-part distribution. We relate $C_{\mathrm{g}}$ to the Mellin transforms $G$ and $A$, derive a geometrically convergent series in even zeta values, and record an enclosure fixing the first 100 decimal digits of $C_{\mathrm{g}}$.

## 1. INTRODUCTION

For integers $n \ge 2$ put

$$P(n) = \prod_{k=2}^{n-1} \sin\left(\frac{\pi n}{k}\right), \qquad \widetilde{P}(n) = \prod_{2\le k\le\sqrt{n}} \sin\left(\frac{\pi n}{k}\right). \tag{1.1}$$

The empty products are $P(2)$, $\widetilde{P}(2)$ and $\widetilde{P}(3)$, each with value 1 by convention. A factor vanishes exactly when $k$ divides $n$, and every composite $n$ has a divisor in $[2,\sqrt{n}]$. Thus each product is nonzero exactly when $n$ is prime. For primes $p$ the logarithm of the absolute value of the product has the asymptotic form

$$\log|P(p)| = -C_{\mathrm{g}}p + O(p^{1/3}\log^2 p), \qquad C_{\mathrm{g}} = 0.74379486183940054420\ldots. \tag{1.2}$$

Consequently $|P(p)|^{1/p} \to e^{-C_{\mathrm{g}}} = 0.47530676524152038801\ldots$. The logarithmic decay rate is $C_{\mathrm{g}}$, and $e^{-C_{\mathrm{g}}}$ is the exponential base. For the product truncated at the square root,

$$\log|\widetilde{P}(p)| = -(\log 2)\sqrt{p} + O(p^{1/3}\log^2 p), \qquad |\widetilde{P}(p)|^{1/\sqrt{p}} \longrightarrow \frac{1}{2}. \tag{1.3}$$

The constants in (1.2) and (1.3) arise from different distributions of the fractional parts.

Let $U$ be uniform on $(0,1)$ and put $X = \{1/U\}$. This is the limiting distribution of the fractional parts $\{n/k\}$ for the full product $P(n)$. The mean of $X$ is $1-\gamma$, where $\gamma$ is Euler's constant, and the mean of $-\log(\sin \pi X)$ is $C_{\mathrm{g}}$. These two means occur in a single complex expectation:

$$\mathbb{E}\,\mathrm{Log}(1 - e^{2\pi i X}) = (\log 2 - C_{\mathrm{g}}) + i\pi\left(\frac{1}{2} - \gamma\right). \tag{1.4}$$

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Here Log is the principal logarithm. This identity is the sense in which $C_{\mathrm{g}}$ is a log-sine analogue of Euler's constant.

For real $u \notin \mathbb{Z}$ put

$$h(u) = \log|2\sin(\pi u)|. \tag{1.5}$$

The transform of $h$ on $[1,\infty)$ is

$$G(s) = \int_1^\infty h(u)\, u^{-s-1}\, du \qquad (\operatorname{Re} s > 0). \tag{1.6}$$

There is also a parallel at the level of Mellin transforms. Put

$$\mathcal{M}_{\mathrm{frac}}(s) = \int_1^\infty \{u\}\, u^{-s-1}\, du \qquad (\operatorname{Re} s > 0).$$

For $\operatorname{Re} s > 1$, integrating $\lfloor u \rfloor = \sum_{n \le u} 1$ termwise gives

$$\zeta(s) = s\int_1^\infty \lfloor u \rfloor u^{-s-1}\, du = \frac{s}{s-1} - s\mathcal{M}_{\mathrm{frac}}(s).$$

The integral defining $\mathcal{M}_{\mathrm{frac}}$ is holomorphic on $\operatorname{Re} s > 0$ because $\{u\}$ is bounded and the integral converges locally uniformly there. The formula thus continues $\zeta$ meromorphically to that half-plane. Comparing constant terms at $s = 1$ with the Laurent expansion of $\zeta$ [3, (25.2.4)–(25.2.5)] gives $\mathcal{M}_{\mathrm{frac}}(1) = 1 - \gamma$. Likewise, $\log(2)/s - G(s)$ is the transform of $-\log|\sin(\pi u)|$ on $[1,\infty)$ and takes the value $C_{\mathrm{g}}$ at $s = 1$ by (3.2). Both values are obtained with the weight $u^{-2}$.

For $0 < \operatorname{Re} s < 2$ let

$$A(s) = \frac{\log(2\pi)}{s} + \frac{1}{s^2} - \int_0^1 \log\frac{\sin(\pi u)}{\pi u}\, u^{-s-1}\, du. \tag{1.7}$$

In particular $C_{\mathrm{g}} = \log 2 - G(1)$. As an identity between meromorphic functions,

$$G(s) - A(s) = \frac{\pi}{s}\cot\Big(\frac{\pi s}{2}\Big)\zeta(s). \tag{1.8}$$

For non-real $s$, equality $G(s) = A(s)$ is equivalent to $\zeta(s) = 0$, with the same multiplicities. Thus if every non-real zero of $\zeta$ has real part $\frac{1}{2}$, the non-real solutions of $G = A$ are of the form $\frac{1}{2} + it$ with $t \neq 0$. The real solutions are the odd integers other than 1. At $s = 1$ the zero of the cotangent cancels the pole of $\zeta$, giving $G(1) - A(1) = -\pi^2/2$.

## 2. Notation and constants

Throughout, $s = \sigma + it$ with $\sigma, t$ real. The index $n$ is an integer, $p$ is a prime, and $k$ and $m$ are summation indices. The fractional part of $x$ is $\{x\} = x - \lfloor x \rfloor$. We use $y$ for a variable in $(0,1)$ in the fractional-part distribution. All real logarithms are natural logarithms.

| Symbol | Meaning | Initial decimals |
|---|---|---|
| $\gamma$ | Euler's constant | $0.57721566490153286060\ldots$ |
| $C_{\mathrm{g}}$ | Logarithmic decay rate of $\lvert P(p)\rvert$ | $0.74379486183940054420\ldots$ |
| $e^{-C_{\mathrm{g}}}$ | Exponential base of $\lvert P(p)\rvert$ | $0.47530676524152038801\ldots$ |
| $\log 2$ | Logarithmic decay rate of $\lvert\widetilde{P}(p)\rvert$ | $0.69314718055994530941\ldots$ |
| $G(1)$ | $\log 2 - C_{\mathrm{g}}$ | $-0.05064768127945523478\ldots$ |
| $A(1)$ | $G(1) + \pi^2/2$ | $4.88415451926522407463\ldots$ |

Table 1. The constants used in the product and its Mellin representation.

Table 2: Notation used in the proofs and appendices.

| Symbol | Meaning |
|---|---|
| $n, p, k, m, j, b, l, e_q$ | Integer arguments and indices. In the divisor bound $q$ is prime and $e_q$ is its exponent in $n$. |
| $P, \widetilde{P}, d(n), Y$ | Full product, square-root product, divisor count, and cutoff in the counting estimate. |
| $U, X, y, y_k$ | Uniform variable on $(0,1)$, its image $\{1/U\}$, a fractional-part variable, and $\{n/k\}$. |
| $w, F$ | Density of $X$ and its distribution function. |
| $\mu, \nu, W, T, D_\mu, \tau$ | Sample measure, signed discrepancy measure, comparison density, comparison mass, uniform discrepancy, and counting threshold. |
| $f, f_\delta, \delta, I, m_0$ | Logarithmic observable, its clipped form, endpoint cutoff, uniform integral of $f$, and a finite sum or product endpoint. |
| $\gamma, C_{\mathrm{g}}, e^{-C_{\mathrm{g}}}, \log 2$ | Euler's constant, the full-product decay rate, its exponential base, and the square-root decay rate. |
| $s, \sigma, t, i, u, v, x, z, \xi$ | Complex argument, its real and imaginary coordinates, the imaginary unit, and local integration and function arguments. In the principal-value calculation $z = s/2$. |
| $\log, \mathrm{Log}, \mathrm{PV}$ | Real natural logarithm, principal complex logarithm, and Cauchy principal value. |
| $h, \ell, G, A$ | Log-sine kernel, regularized logarithm, and the two functions in (1.6) and (1.7). |
| $\mathcal{M}_{\mathrm{frac}}$ | Mellin transform of the fractional part on $[1,\infty)$. |
| $\zeta, \Gamma, \Sigma_{2k}, J(s), \psi$ | Zeta and gamma functions, even zeta values, the scalar Mellin integral, and the digamma function. |
| $N_C, C_{\mathrm{g}}^{[N_C]}, B_{2k}$ | Number of terms in the series for $C_{\mathrm{g}}$, its partial value, and the even Bernoulli numbers. |
| $M_C, a, R_{N_C,M_C}$ | Cutoff in the elementary series, the first omitted index $a = M_C + 1$, and the remainder of the accelerated formula. |
| $D_{100}, L_{180}, U_{180}$ | Integer formed by the first 100 fractional digits, and the endpoints enclosing the 180-term partial value. |
| $\mathcal{K}, \mathcal{K}_\alpha, H$ | Periodic kernel, its Abel regularization with $0 < \alpha < 1$, and its Mellin integral. |
| $\mathsf{M}, \mathsf{N}, \mathsf{L}, x_b$ | Numbers of harmonics, periods, and midpoint nodes in the Rouché certificate, and its midpoint nodes. |
| $\mathcal{K}_{\mathsf{M}}, H_{\mathsf{M},\mathsf{N}}, H_{\mathsf{M},\mathsf{N},\mathsf{L}}$ | Kernel cutoff, cutoff integral, and midpoint approximation. |
| $B_0, B_1, B_2, Z, E_0, E_1$ | Bounds for the kernel and its derivatives, an upper bound for a sum of inverse powers, and the errors for $H$ and $H'$. |
| $s_0, \varsigma, \eta_0, \eta_1$ | Disc center and radius, and the approximations to $H(s_0)$ and $H'(s_0)$. |
| $Q_0, \mathcal{Q}, \mathfrak{r}_k, c_k, g_\psi$ | Digamma remainder series, upper bound for $\mathrm{Re}(sA(s))$, $\mathrm{Re}(s/(2k-s))$, $\Sigma_{2k}-1$, and the nonnegative digamma comparison integrand. |

| Symbol | Meaning |
|---|---|
| $\varepsilon, \rho$ | Exponent in the divisor bound and cutoff for the principal value. |
| $u_0, \theta, \lambda, \chi$ | Tail endpoint, inverse-power exponent, local integration variable, and midpoint-rule integrand. |
| $g_{jk}, r, \beta$ | Greatest common divisor $\gcd(j,k)$, positive integer multiple, and frequency in the Gram calculation. |
| $z_b^G, z_b^A, n_G, n_A, \mathcal{E}_G, \mathcal{E}_A$ | Lists of computed zeros, their lengths, and the exponential sums. |

## 3. The distribution and the arithmetic product

Splitting $1/U$ into intervals $[m, m+1)$ gives the density

$$w(y) = \sum_{m=1}^{\infty} \frac{1}{(m+y)^2}, \qquad 0 < y < 1. \tag{3.1}$$

Indeed the change of variables $u = 1/(m+y)$ on each interval of $U$ contributes $(m+y)^{-2}\,dy$. The sum of the integrals of these terms is 1. This distribution for $\{n/k\}$ is treated by Saffari and Vaughan [5, Corollary 1.1] and Bensimhoun [2, Examples 3–4]. Termwise integration of $yw(y)$ up to the summand $m = m_0$ gives

$$\sum_{m=1}^{m_0} \left( \log \frac{m+1}{m} - \frac{1}{m+1} \right) = 1 + \log(m_0+1) - \sum_{m=1}^{m_0+1} \frac{1}{m}.$$

Its limit is $1 - \gamma$. Integrating the log-sine observable gives

$$C_{\mathrm{g}} = -\int_0^1 \log(\sin \pi y)\, w(y)\, dy = -\int_1^{\infty} \frac{\log|\sin(\pi u)|}{u^2}\, du. \tag{3.2}$$

The first integral converges because $w$ is bounded and the endpoint singularities are logarithmic. The second equality follows by splitting the integral on $[1, \infty)$ into unit intervals. For $0 < y < 1$ the identity $1 - e^{2\pi i y} = 2\sin(\pi y) e^{i\pi(y-1/2)}$ has argument in $(-\pi/2, \pi/2)$. Taking the mean of its principal logarithm gives (1.4).

**Lemma 3.1** (Discrepancy and the logarithmic endpoints)**.** *Let $n \geq 3$, and let $\mu$ count a finite list of points in $[1/n, 1-1/n]$, with multiplicities. Let $W$ be a bounded nonnegative density of integral 1 on $[0,1]$, and let $T > 0$. Put*

$$D_\mu = \sup_{0 \leq \tau \leq 1} \left| \mu([0,\tau]) - T \int_0^{\tau} W(y)\, dy \right|.$$

*For $f(y) = \log(\sin \pi y)$ and $1/n \leq \delta < 1/2$,*

$$\int f\, d\mu = T \int_0^1 f(y) W(y)\, dy + O((D_\mu + T\delta) \log n),$$

*with an implied constant depending only on an upper bound for $W$.*

*Proof.* Define $f_\delta = f$ on $[\delta, 1-\delta]$ and extend it constantly to each endpoint of $[0,1]$. Since $f_\delta(1) = f(1-\delta) = \log(\sin \pi\delta)$, the inequality $\sin(\pi y) \geq 2\min(y, 1-y)$ and the monotonicity of $f$ on each side of $1/2$ give

$$|f_\delta(1)| + \int_0^1 |f_\delta'(y)|\, dy = O(\log n).$$

For $\nu = \mu - TW(y)\,dy$, insert $f_\delta(y) = f_\delta(1) - \int_y^1 f_\delta'(\tau)\,d\tau$ in $\int f_\delta\,d\nu$ and interchange the finite-measure integrals. The result is bounded in modulus by $D_\mu(|f_\delta(1)| + \int_0^1 |f_\delta'|)$. The two endpoint intervals contain $O(T\delta + D_\mu)$ sample points by the definition of $D_\mu$ and boundedness of $W$. Both $f$ and $f_\delta$ have size $O(\log n)$ at those points, since the sample lies in $[1/n, 1-1/n]$ and $\delta \geq 1/n$. On the integral side, $\int_0^\delta |\log(2y)|\,dy = O(\delta(1 + |\log\delta|))$, with the same bound at the other endpoint by reflection. Thus replacing $f_\delta$ by $f$ costs $O((T\delta + D_\mu)\log n)$ in the sample sum and $O(T\delta \log n)$ in the integral against $TW(y)\,dy$. These are the claimed bounds. □

**Theorem 3.2** (Arithmetic sampling)**.** *Let $d(n)$ be the number of positive divisors of $n$. For every integer $n \geq 2$, with absolute implied constants,*

$$\sum_{\substack{2\leq k\leq n-1\\ k\nmid n}} \log\left|\sin\frac{\pi n}{k}\right| = -C_{\mathrm{g}} n + O\big(n^{1/3}\log^2 n + d(n)\log n\big), \tag{3.3}$$

$$\sum_{\substack{2\leq k\leq \sqrt{n}\\ k\nmid n}} \log\left|\sin\frac{\pi n}{k}\right| = -(\log 2)\sqrt{n} + O\big(n^{1/3}\log^2 n + d(n)\log n\big). \tag{3.4}$$

*Proof.* For the full range, let $F(\tau) = \int_0^\tau w(y)\,dy = \sum_{m\geq 1} \tau/[m(m+\tau)]$ for $0 \leq \tau \leq 1$. Saffari and Vaughan [5, Corollary 1.1, equation (1.9)] give

$$\#\{1 \leq k \leq n : \{n/k\} < \tau\} = nF(\tau) + O(n^{1/3}\log n) \qquad (0 \leq \tau \leq 1)$$

uniformly in $\tau$. The points at zero are the $d(n)$ divisors of $n$, including 1 and $n$. Passing to right limits in the counting estimate and removing the divisors gives $D_\mu = O(n^{1/3}\log n + d(n))$ for $T = n$ and $W = w$ in Lemma 3.1. The endpoint $\tau = 1$ follows from the count of the remaining $n - d(n)$ points.

For the square-root range, Saffari and Vaughan [5, Corollary 1.3, equation (1.11)] give, for $Y = o(n)$, the uniform bound

$$\#\{1 \leq k \leq Y : \{n/k\} < \tau\} = Y\tau + O(Y^2/n + n^{1/3}\log n) \qquad (0 \leq \tau \leq 1).$$

With $Y = \sqrt{n}$ the first error term is 1. As before, taking right limits and removing the divisors, including 1, gives $D_\mu = O(n^{1/3}\log n + d(n))$, this time for $T = \sqrt{n}$ and $W = 1$. At $\tau = 1$ the remaining count is

$$\mu([0,1]) = \lfloor\sqrt{n}\rfloor - \#\{k : 1 \leq k \leq \sqrt{n},\ k \mid n\}.$$

Thus $|\mu([0,1]) - \sqrt{n}| \leq d(n) + 1$. The bounds also give the distributional limits in each range, since $D_\mu/T \to 0$. For the latter assertion one may use the elementary divisor bound $d(n) = O_\varepsilon(n^\varepsilon)$ for any $\varepsilon > 0$. Indeed, if $n = \prod_q q^{e_q}$, then $e_q + 1 \leq 2^{e_q} \leq q^{\varepsilon e_q}$ for $q \geq 2^{1/\varepsilon}$. For each of the finitely many smaller primes $q$, the ratio $(e_q+1)/q^{\varepsilon e_q}$ is bounded over integers $e_q \geq 0$. Multiplication gives the bound.

In either range $y_k = \{n/k\}$ for $k \nmid n$ satisfies $\min(y_k, 1-y_k) \geq 1/k \geq 1/n$. Writing $n/k = \lfloor n/k\rfloor + y_k$ gives $\sin(\pi n/k) = (-1)^{\lfloor n/k\rfloor}\sin(\pi y_k)$, and hence $\log|\sin(\pi n/k)| = \log(\sin(\pi y_k))$ because $0 < y_k < 1$. Use Lemma 3.1 with $\delta = n^{-2/3}$ for $T = n$, and with $\delta = n^{-1/6}$ for $T = \sqrt{n}$. In both cases $T\delta = n^{1/3}$. The finitely many $n$ for which the prescribed $\delta$ is at least $1/2$ are covered by enlarging the implied constant. The full-range integral is $-C_{\mathrm{g}}$ by (3.2). In the square-root range write $I = \int_0^1 \log(\sin\pi y)\,dy$. By a shift of half a period,

$\int_0^1 \log|\cos(\pi y)|\,dy = I$. The double-angle formula then gives $2I = I - \log 2$, because $\int_0^1 \log|\sin(2\pi y)|\,dy = I$. This gives $I = -\log 2$ and proves (3.4). □

For $n = p$ prime, $d(p) = 2$ and neither product loses any factors in the respective sums. Since $2\log p = O(p^{1/3}\log^2 p)$, equation (3.3) gives (1.2), while (3.4) gives the logarithmic estimate in (1.3). Division by $\sqrt{p}$ and exponentiation give the limit $1/2$ in (1.3).

## 4. The Mellin representation and the zeta series

For real $u > 0$ with $u \notin \mathbb{Z}$ put

$$\ell(u) = \log \frac{\pi u}{|\sin(\pi u)|}. \tag{4.1}$$

By (1.5), $h(u) - \log(2\pi u) = -\ell(u)$. The definitions (1.6) and (1.7) then give

$$G(s) - A(s) = -\int_0^\infty \ell(u)\, u^{-s-1}\,du \qquad (0 < \operatorname{Re} s < 2). \tag{4.2}$$

To see this, split the integral at 1 and use $\int_1^\infty \log(2\pi u)u^{-s-1}\,du = \log(2\pi)/s + 1/s^2$. Near zero the logarithm is $O(u^2)$. At infinity it is the sum of a logarithm and a periodic function with integrable singularities. More explicitly, $\int_m^{m+1}|h(u)|\,du = \int_0^1 |h(u)|\,du < \infty$. These bounds give absolute convergence, locally uniformly for $0 < \operatorname{Re} s < 2$, and show that (1.6) is holomorphic for $\operatorname{Re} s > 0$. In the notation of Adam [1, Theorem 2.2, definition of $I_s$], the specialization $(1, 0, \frac{1}{4}, 1)$ gives, for $\operatorname{Re} s > 0$,

$$I_s(1, 0, \tfrac{1}{4}, 1) = -\frac{1}{2}\int_1^\infty \log\bigl(4\sin^2(\pi u)\bigr)u^{-s-1}\,du = -G(s).$$

In particular, $C_{\mathrm{g}} = \log 2 + I_1(1, 0, \frac{1}{4}, 1)$.

**Theorem 4.1** (Continuation and Mellin identity)**.** *The function $A$ extends meromorphically to $\mathbb{C}$ by the series*

$$A(s) = \frac{\log(2\pi)}{s} + \frac{1}{s^2} + \sum_{k\ge 1} \frac{\Sigma_{2k}}{k\,(2k - s)}, \tag{4.3}$$

*where $\Sigma_{2k} = \sum_{m\ge1} m^{-2k} = \zeta(2k)$. The function $G$ extends to an entire function and* (1.8) *holds meromorphically on $\mathbb{C}$.*

*Proof.* Euler's product for the sine [3, (4.22.1)] gives, for $0 < u < 1$,

$$-\log\frac{\sin(\pi u)}{\pi u} = \sum_{m\ge1}\sum_{k\ge1}\frac{u^{2k}}{km^{2k}} = \sum_{k\ge1}\frac{\Sigma_{2k}}{k}u^{2k}.$$

For $\sigma < 2$ the sum of the integrals of the absolute values after multiplication by $u^{-s-1}$ is $\sum_{k\ge1}\Sigma_{2k}/[k(2k-\sigma)] < \infty$. Integration therefore proves (4.3). The large-$k$ terms are $O(k^{-2})$ uniformly on each compact set, which gives the claimed meromorphic continuation. Its only poles are a double pole at 0 and simple poles at the positive even integers.

For positive nonintegral $u$, the same sine product gives $-\ell(u) = \sum_{m\ge1}\log|1 - u^2/m^2|$. If $1 < \sigma < 2$, the sum of the weighted absolute integrals on $(0,\infty)$ is

$$\sum_{m\ge1} m^{-\sigma}\int_0^\infty |\log|1 - v^2||\,v^{-\sigma-1}\,dv < \infty.$$

The inner integral is finite by the $O(v^2)$ behavior at zero, the logarithmic singularity at 1, and $O(\log v)$ growth at infinity. Termwise integration in (4.2), with $u = mv$, now gives $G(s) - A(s) = \zeta(s)J(s)$, where

$$J(s) = \int_0^\infty \log|1 - v^2|\, v^{-s-1}\, dv = \frac{1}{2}\int_0^\infty \log|1-\xi|\, \xi^{-s/2-1}\, d\xi.$$

Write $z = s/2$. Integration by parts on the intervals separated by $1-\rho$ and $1+\rho$ gives $J(s) = s^{-1}\,\mathrm{PV}\int_0^\infty \xi^{-z}/(\xi-1)\,d\xi$. The boundary terms at zero and infinity vanish for $0 < \operatorname{Re} z < 1$, and the paired terms at $1 \pm \rho$ are $O(\rho|\log\rho|)$. Substitution $\xi = 1/y$ above 1 changes the cutoff there to $1/(1+\rho)$. Replacing it by $1-\rho$ changes the integral by $O(\rho)$, since the endpoints are $O(\rho^2)$ apart and the integrand there is $O(\rho^{-1})$. The principal value is

$$\int_0^1 \frac{y^{z-1} - y^{-z}}{1-y}\, dy = \sum_{m=0}^{\infty}\left(\frac{1}{m+z} - \frac{1}{m+1-z}\right) = \pi\cot(\pi z).$$

The geometric-series integration is justified by the integrability of the modulus of the difference divided by $1-y$, since the numerator vanishes at 1. The last equality is the partial fraction expansion of the cotangent [3, (4.22.3)]. This proves (1.8) on $1 < \operatorname{Re} s < 2$.

Define the continuation of $G$ by adding the right side of (1.8) to (4.3). At a positive even integer $2k$, the residues of the two summands are $-\zeta(2k)/k$ and $\zeta(2k)/k$. At a negative even integer the zero of $\zeta$ [3, (25.6.4)] cancels the cotangent pole. The factor $\cot(\pi s/2)$ cancels the pole of $\zeta$ at 1. At zero, $(\pi/s)\cot(\pi s/2) = 2/s^2 + O(1)$ and $\zeta(0) = -1/2$, $\zeta'(0) = -\log(2\pi)/2$ [3, (25.6.1) and (25.6.11)]. The two principal parts at zero cancel. There are no other possible poles. This entire function agrees with the integral definition of $G$ on $1 < \operatorname{Re} s < 2$. Since both functions are holomorphic on the connected half-plane $\operatorname{Re} s > 0$, the identity theorem shows that they agree throughout $\operatorname{Re} s > 0$. □

**Proposition 4.2.** *For every non-real $s$, $G(s) = A(s)$ if and only if $\zeta(s) = 0$, and the multiplicity of $s$ as a zero of $G - A$ equals its multiplicity as a zero of $\zeta$. The real zeros of $G - A$ are simple and occur at the odd integers different from* 1.

*Proof.* For non-real $s$ the factor $(\pi/s)\cot(\pi s/2)$ in (1.8) is holomorphic and nonzero, so it preserves zeros and their multiplicities. For real $s < 0$, the functional equation [3, (25.4.2)] gives, after cancellation of the sine factor,

$$G(s) - A(s) = \frac{(2\pi)^s}{s}\cos\frac{\pi s}{2}\,\Gamma(1-s)\zeta(1-s).$$

The equality holds at the negative even integers by continuation. Since $\Gamma(1-s)$ and $\zeta(1-s)$ are positive and finite on $s < 0$, the only zeros there are the simple cosine zeros at the negative odd integers.

For $0 < s < 1$, (4.2) gives $G(s) - A(s) < 0$, since $|\sin(\pi u)| < \pi u$ for $u > 0$. For $s > 1$, the Dirichlet series gives $\zeta(s) > 0$. The cotangent factor then gives simple zeros at $3, 5, \ldots$ and poles at the positive even integers. This covers all the real nonintegers as well. At 0 there is a double pole, since $\zeta(0) = -1/2$ and $(\pi/s)\cot(\pi s/2) = 2/s^2 + O(1)$. At 1, $(s-1)\zeta(s) \to 1$ and $\cot(\pi s/2) = -\pi(s-1)/2 + O((s-1)^3)$. Hence the value at 1 is $-\pi^2/2$. □

**Theorem 4.3.** *The constant $C_{\mathrm{g}}$ satisfies*

$$C_{\mathrm{g}} = \frac{\pi^2}{2} - \log(2\pi) - 1 - 2\sum_{k=1}^{\infty} \frac{\zeta(2k)-1}{2k-1}. \tag{4.4}$$

*If $C_{\mathrm{g}}^{[N_C]}$ denotes the right side with $1 \le k \le N_C$, then*

$$\frac{2(\zeta(2N_C+2)-1)}{2N_C+1} \le C_{\mathrm{g}}^{[N_C]} - C_{\mathrm{g}} \le \frac{2\cdot 4^{-N_C}}{2N_C+1} \qquad (N_C \ge 1). \tag{4.5}$$

*Proof.* Equation (3.2) gives $C_{\mathrm{g}} = \log 2 - G(1)$. The value of (1.8) at 1 gives $G(1) = A(1) - \pi^2/2$. Substitution of (4.3) therefore yields

$$C_{\mathrm{g}} = \frac{\pi^2}{2} - \log\pi - 1 - \sum_{k\ge 1} \frac{\zeta(2k)}{k(2k-1)}.$$

The finite product $\prod_{m=2}^{m_0}(1-m^{-2}) = (m_0+1)/(2m_0)$ gives $\sum_{k\ge1}(\zeta(2k)-1)/k = \log 2$ by summing the logarithmic series, whose terms are nonnegative. Also $\sum_{k\ge1} 1/[k(2k-1)] = 2\log 2$ by the alternating harmonic series. Using $1/[k(2k-1)] = 2/(2k-1) - 1/k$ gives (4.4). For the remainder,

$$0 < \zeta(2k) - 1 \le 4^{-k} + \int_2^{\infty} v^{-2k}\, dv \le 3\cdot 4^{-k}.$$

For $k > N_C$ we have $2k-1 \ge 2N_C+1$ and $\sum_{k>N_C} 3\cdot 4^{-k} = 4^{-N_C}$, which gives the upper bound. Keeping the first positive term of the remainder gives the lower bound. Since $\zeta(2N_C+2) - 1 \ge 4^{-(N_C+1)}$, the remainder has order $4^{-N_C}/N_C$ as $N_C \to \infty$. □

**Proposition 4.4** (Elementary series and acceleration)**.** *The constant also satisfies*

$$C_{\mathrm{g}} = \frac{\pi^2}{2} - \log(2\pi) - 1 - \sum_{m\ge2} \frac{1}{m}\log\frac{m+1}{m-1}. \tag{4.6}$$

*For any integer $M_C \ge 2$, put $a = M_C + 1$. Then*

$$\begin{aligned} C_{\mathrm{g}} = \frac{\pi^2}{2} - \log(2\pi) - 1 &- \sum_{m=2}^{M_C} \frac{1}{m}\log\frac{m+1}{m-1} \\ &- 2\sum_{k\ge1} \frac{\zeta(2k,a)}{2k-1}, \end{aligned} \tag{4.7}$$

*where $\zeta(s,a)$ is the Hurwitz zeta function. Truncating the last sum at $k = N_C$ gives an overestimate with remainder $R_{N_C,M_C}$ satisfying*

$$\frac{2\zeta(2N_C+2,a)}{2N_C+1} \le R_{N_C,M_C} \le \frac{2\zeta(2N_C+2,a)}{(2N_C+1)(1-a^{-2})}. \tag{4.8}$$

*Proof.* All terms in the double series below are nonnegative. Tonelli's theorem and the series for artanh give

$$\begin{aligned} 2\sum_{k\ge1} \frac{\zeta(2k)-1}{2k-1} &= 2\sum_{m\ge2}\sum_{k\ge1} \frac{m^{-2k}}{2k-1} \\ &= 2\sum_{m\ge2} \frac{1}{m}\operatorname{artanh}\frac{1}{m} = \sum_{m\ge2}\frac{1}{m}\log\frac{m+1}{m-1}. \end{aligned}$$

The summand is $2m^{-2} + O(m^{-4})$. Keeping the terms $2 \le m \le M_C$ in this form and using $\zeta(2k, a) = \sum_{m \ge a} m^{-2k}$ [3, (25.11.1)] for the remaining terms proves (4.7). The first omitted term gives the lower bound in (4.8). For the upper bound use $\zeta(2k+2, a) \le a^{-2}\zeta(2k, a)$ and sum a geometric series, bounding all the denominators below by $2N_C + 1$. Finally, for $k \ge 1$, the integral test gives

$$a^{-2k} \le \zeta(2k, a) \le a^{-2k}\left(1 + \frac{a}{2k-1}\right).$$

Thus for each fixed $M_C$ the remainder has order $(M_C+1)^{-2N_C}/N_C$ as $N_C \to \infty$. □

## References


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## Appendix A. Numerical certificates

A.1. **Decimal enclosure for $C_{\mathrm{g}}$.** The computation retained $N_C = 180$ terms of (4.4) in 150-decimal interval arithmetic. The even zeta values were enclosed using Euler's formula [3, (25.6.2)],

$$\zeta(2k) = \frac{|B_{2k}|(2\pi)^{2k}}{2(2k)!}$$

with exact rational Bernoulli numbers $B_{2k}$. If $[L_{180}, U_{180}]$ encloses the partial sum, the computation used the weaker error bound $2 \cdot 4^{-180}$ implied by (4.5). Thus

$$C_{\mathrm{g}} \in [L_{180} - 2 \cdot 4^{-180}, U_{180}].$$

The two endpoints, multiplied by $10^{100}$, have the same integer part $D_{100}$, whose digits are printed below. Thus $D_{100}\,10^{-100} \le C_{\mathrm{g}} < (D_{100}+1)10^{-100}$.

$$\begin{aligned} C_{\mathrm{g}} = 0.&7437948618\,3940054420\,3641994634\,5806610252\,6433197291 \\ &7222960354\,3808135540\,0286772998\,3871757037\,1013522332\ldots. \end{aligned} \tag{A.1}$$

The recorded result and the calculation script are `certificate_Cg_v1.0.txt` and `verify_Cg_v1.0.py` in the accompanying files.

A.2. **Kernel and Rouché test for $G$.** Put

$$\mathcal{K}(u) = \sum_{m\ge 1} \frac{1-\cos(2\pi m u)}{m^3}. \tag{A.2}$$

Its Mellin integral is

$$H(s) = \int_1^\infty \mathcal{K}(u) u^{-s-3}\,du \quad (\operatorname{Re} s > -2). \tag{A.3}$$

The kernel series converges uniformly, and $0 \le \mathcal{K} \le 2\zeta(3) < 5/2$ by the integral test starting at $m = 2$. In particular $H$ is holomorphic on $\operatorname{Re} s > -2$.

To relate $H$ to $G$, insert a factor $\alpha^m$ in the kernel series for $0 < \alpha < 1$ and denote the resulting kernel by $\mathcal{K}_\alpha$. The logarithmic series at $\alpha e^{2\pi i u}$ gives $\mathcal{K}_\alpha''(u) = -4\pi^2 \log|1-\alpha e^{2\pi i u}|$. Integration by parts twice against $u^{-s-1}$ on $[1,\infty)$ for $\operatorname{Re} s > 0$ has zero boundary terms, since $\mathcal{K}_\alpha(1) = \mathcal{K}_\alpha'(1) = 0$ and the kernel and its derivative are bounded. To let $\alpha \uparrow 1$, use uniform convergence of $\mathcal{K}_\alpha$ to $\mathcal{K}$ and, for $1/2 \le \alpha < 1$, the bounds

$$2\sin^2(\pi u) \le |1-\alpha e^{2\pi i u}|^2 = (1-\alpha)^2 + 4\alpha \sin^2(\pi u) \le 4.$$

They give a periodic integrable majorant for the logarithm on each period and hence a weighted integrable majorant on $[1,\infty)$. We obtain

$$G(s) = -\frac{(s+1)(s+2)}{4\pi^2} H(s) \qquad (\operatorname{Re} s > -2), \tag{A.4}$$

where continuation to $\operatorname{Re} s > -2$ follows by the identity theorem.

Let $\mathsf{M}, \mathsf{N}, \mathsf{L} \ge 1$ be integers and $x_b = (b+\frac{1}{2})/\mathsf{L}$ for $0 \le b \le \mathsf{L}-1$. Put

$$\mathcal{K}_{\mathsf{M}}(x) = \sum_{m=1}^{\mathsf{M}} \frac{1-\cos(2\pi m x)}{m^3}, \qquad H_{\mathsf{M},\mathsf{N}}(s) = \sum_{n=1}^{\mathsf{N}} \int_0^1 \mathcal{K}_{\mathsf{M}}(x)\,(n+x)^{-s-3}\,dx, \tag{A.5}$$

$$H_{\mathsf{M},\mathsf{N},\mathsf{L}}(s) = \frac{1}{\mathsf{L}} \sum_{n=1}^{\mathsf{N}} \sum_{b=0}^{\mathsf{L}-1} \mathcal{K}_{\mathsf{M}}(x_b)\,(n+x_b)^{-s-3}. \tag{A.6}$$

Put also

(A.7)

$$B_0 = \frac{5}{2}, \qquad B_1 = \frac{\pi^3}{3}, \qquad B_2 = 4\pi^2(1+\log \mathsf{M}), \qquad Z(\theta) = 1 + \frac{1}{\theta - 1} \quad (\theta > 1).$$

For $\theta > 1$ we have $\sum_{n\ge 1} n^{-\theta} \le 1 + \int_1^\infty x^{-\theta} dx = Z(\theta)$. The functions $H_{\mathsf{M},\mathsf{N}}$ and $H_{\mathsf{M},\mathsf{N},\mathsf{L}}$ are entire. By (A.5), $0 \le \mathcal{K}_{\mathsf{M}} \le B_0$, $|\mathcal{K}'_{\mathsf{M}}| \le 2\pi \sum_{m\ge 1} m^{-2} = B_1$ and $|\mathcal{K}''_{\mathsf{M}}| \le 4\pi^2 \sum_{m\le \mathsf{M}} 1/m \le B_2$. We use the midpoint rule in the following form. If $\chi$ is a twice continuously differentiable complex function on $[0,1]$, then

$$\Big| \int_0^1 \chi(x)\,dx - \frac{1}{\mathsf{L}} \sum_{b=0}^{\mathsf{L}-1} \chi(x_b) \Big| \le \frac{\sup |\chi''|}{24\,\mathsf{L}^2}. \tag{A.8}$$

Indeed, on the subinterval of length $1/\mathsf{L}$ centred at $x_b$, Taylor's formula gives $|\chi(x) - \chi(x_b) - \chi'(x_b)(x-x_b)| \le \frac{1}{2} \sup|\chi''| (x-x_b)^2$, the linear term integrates to 0, and $\int (x-x_b)^2 dx = 1/(12\mathsf{L}^3)$ over that subinterval.

**Theorem A.1** (Explicit error bounds)**.** *Let $\sigma = \operatorname{Re} s > -1$. Then $|H(s) - H_{\mathsf{M},\mathsf{N},\mathsf{L}}(s)| \le E_0$ and $|H'(s) - H'_{\mathsf{M},\mathsf{N},\mathsf{L}}(s)| \le E_1$, where*

$$\begin{aligned} E_0 = {} & \frac{1}{(\sigma+2)\mathsf{M}^2} + \frac{B_0}{(\sigma+2)(\mathsf{N}+1)^{\sigma+2}} \\ & + \frac{B_2 Z(\sigma+3) + 2|s+3| B_1 Z(\sigma+4) + |(s+3)(s+4)| B_0 Z(\sigma+5)}{24\,\mathsf{L}^2} \end{aligned} \tag{A.9}$$

*and*

$$\begin{aligned} E_1 = {} & \frac{1}{(\sigma+2)^2\mathsf{M}^2} + \frac{B_0}{(\mathsf{N}+1)^{\sigma+2}} \Big( \frac{\log(\mathsf{N}+1)}{\sigma+2} + \frac{1}{(\sigma+2)^2} \Big) \\ & + \frac{B_2 Z(\sigma+2) + 2B_1 \big( Z(\sigma+4) + |s+3| Z(\sigma+3) \big)}{24\,\mathsf{L}^2} \\ & + \frac{B_0 \big( |(s+3)(s+4)| Z(\sigma+4) + |2s+7| Z(\sigma+5) \big)}{24\,\mathsf{L}^2}. \end{aligned} \tag{A.10}$$

*Proof.* By (A.2) and (A.5), $0 \le \mathcal{K} - \mathcal{K}_{\mathsf{M}} \le 2\sum_{m>\mathsf{M}} m^{-3} \le 2\int_{\mathsf{M}}^\infty x^{-3} dx = \mathsf{M}^{-2}$. The difference $H - H_{\mathsf{M},\mathsf{N}}$ is the integral of $(\mathcal{K} - \mathcal{K}_{\mathsf{M}})(u) u^{-s-3}$ over $1 \le u \le \mathsf{N}+1$ plus the integral of $\mathcal{K}(u) u^{-s-3}$ over $u \ge \mathsf{N}+1$. With $\int_1^\infty u^{-\sigma-3} du = 1/(\sigma+2)$, $\int_{\mathsf{N}+1}^\infty u^{-\sigma-3} du = (\mathsf{N}+1)^{-\sigma-2}/(\sigma+2)$ and $\mathcal{K} \le B_0$, this gives the first two terms of $E_0$. For the third, apply (A.8) to $\chi_n(x) = \mathcal{K}_{\mathsf{M}}(x)(n+x)^{-s-3}$ for $1 \le n \le \mathsf{N}$. Its second derivative is $\mathcal{K}''_{\mathsf{M}}(n+x)^{-s-3} - 2(s+3)\mathcal{K}'_{\mathsf{M}}(n+x)^{-s-4} + (s+3)(s+4)\mathcal{K}_{\mathsf{M}}(n+x)^{-s-5}$, and $|(n+x)^{-s-\theta}| \le n^{-\sigma-\theta}$ for $0 \le x \le 1$ and $\sigma + \theta > 0$. Summing over $n$ with $\sum_n n^{-\theta} \le Z(\theta)$ gives the third term of $E_0$.

For the derivative, the integrand is $-\mathcal{K}(u)(\log u) u^{-s-3}$. Integration gives, for $u_0 \ge 1$,

$$\int_{u_0}^\infty (\log u) u^{-\sigma-3}\,du = u_0^{-\sigma-2} \left( \frac{\log u_0}{\sigma+2} + \frac{1}{(\sigma+2)^2} \right).$$

Taking $u_0 = 1$ and $u_0 = \mathsf{N}+1$ gives the first two terms of $E_1$. For the third, apply (A.8) to $\chi_n(x) = -\mathcal{K}_{\mathsf{M}}(x)\log(n+x)(n+x)^{-s-3}$. For $u \geq 1$,

$$\begin{aligned} \frac{d}{du}\big[(\log u)u^{-s-3}\big] &= u^{-s-4}\big(1-(s+3)\log u\big), \\ \frac{d^2}{du^2}\big[(\log u)u^{-s-3}\big] &= u^{-s-5}\big((s+3)(s+4)\log u-(2s+7)\big). \end{aligned} \tag{A.11}$$

With $0 \leq \log u \leq u$, the moduli of $(\log u)u^{-s-3}$ and of the two derivatives in (A.11) are at most $u^{-\sigma-2}$, $u^{-\sigma-4}+|s+3|u^{-\sigma-3}$ and $|(s+3)(s+4)|u^{-\sigma-4}+|2s+7|u^{-\sigma-5}$. These bounds decrease in $u$ because $\sigma > -1$, so on the $n$-th period they are at most their values at $u = n$. The product rule and $\sum_n n^{-\theta} \leq Z(\theta)$ give the third term of $E_1$. □

**Lemma A.2** (Second derivative)**.** *If* $\operatorname{Re} s \geq \sigma > -2$ *with* $\sigma$ *real, then* $|H''(s)| \leq 2B_0/(\sigma+2)^3$.

*Proof.* By (A.3) and $0 \leq \mathcal{K} \leq B_0$, $|H''(s)| \leq B_0\int_1^\infty (\log u)^2 u^{-\sigma-3}du = 2B_0/(\sigma+2)^3$. □

**Proposition A.3** (Rouché test for a disc)**.** *Let* $s_0 \in \mathbb{C}$ *and* $\varsigma > 0$ *with* $\operatorname{Re} s_0 - \varsigma > -1$. *Let* $\eta_0 = H_{\mathsf{M},\mathsf{N},\mathsf{L}}(s_0)$ *and* $\eta_1 = H'_{\mathsf{M},\mathsf{N},\mathsf{L}}(s_0)$, *and let* $E_0, E_1$ *be the bounds of Theorem A.1 at* $s = s_0$. *If*

$$|\eta_0| + E_0 + \varsigma E_1 + \frac{\varsigma^2 B_0}{(\operatorname{Re} s_0 - \varsigma + 2)^3} < \varsigma\,|\eta_1|, \tag{A.12}$$

*then* $H$ *has exactly one zero in the disc* $|s - s_0| < \varsigma$, *and it is simple. It is also a simple zero of* $G$, *and* $G$ *has no other zero in the disc.*

*Proof.* For $|s-s_0| = \varsigma$, Taylor's formula with integral remainder and Lemma A.2 with $\sigma = \operatorname{Re} s_0 - \varsigma$ give $|H(s) - H(s_0) - H'(s_0)(s-s_0)| \leq \frac{\varsigma^2}{2}\cdot 2B_0/(\operatorname{Re} s_0 - \varsigma + 2)^3$. By Theorem A.1, $|H(s_0) - \eta_0| \leq E_0$ and $|H'(s_0) - \eta_1| \leq E_1$. Hence $|H(s) - \eta_1(s-s_0)|$ is at most the left side of (A.12), which is less than $\varsigma|\eta_1| = |\eta_1(s-s_0)|$. By Rouché's theorem $H$ has in the disc as many zeros, counted with multiplicity, as the function $\eta_1(s-s_0)$, namely one. The disc lies in $\operatorname{Re} s > -1$, where the factor $(s+1)(s+2)$ of (A.4) does not vanish. □

The accompanying Rouché certificate uses Arb [4] at 128 bits with $\mathsf{M} = 4000$, $\mathsf{N} = 200$, $\mathsf{L} = 65536$ and radius $1/100$. Each disc below contains exactly one zero of $G$, counted with multiplicity. The last column gives upper bounds rounded upward from the certified ratios of the two sides of (A.12).

| Disc | $\operatorname{Re} s_0$ | $\operatorname{Im} s_0$ | Rouché ratio, upper bound |
|---|---|---|---|
| 1 | 1.127312874 | 15.001263432 | 0.045205 |
| 2 | 0.533572290 | 21.763254707 | 0.145659 |
| 3 | 0.894730810 | 25.624836921 | 0.155103 |
| 4 | 0.227688021 | 30.975962723 | 0.466950 |
| 5 | 0.950506780 | 33.599087545 | 0.320245 |
| 6 | 0.430899874 | 38.054886210 | 0.452453 |

The interval data and the program and log that produce them accompany the paper.

A.3. **Nonvanishing of $A$.** We use the digamma series [3, (5.7.6)],

$$\psi(z) = -\gamma + \sum_{l\ge 0}\left(\frac{1}{l+1} - \frac{1}{l+z}\right) \qquad (\operatorname{Re} z > 0). \tag{A.13}$$

In (4.3) write $\Sigma_{2k} = 1 + (\Sigma_{2k} - 1)$ and put

$$Q_0(s) = \sum_{k\ge 1} \frac{\Sigma_{2k} - 1}{k(2k-s)} \qquad (\operatorname{Re} s < 2). \tag{A.14}$$

The identity $s/[k(2k-s)] = 1/(k - s/2) - 1/k$ and (A.13) give

$$sA(s) = \log(2\pi) + \frac{1}{s} - \psi\Big(1 - \frac{s}{2}\Big) - \gamma + sQ_0(s) \qquad (\operatorname{Re} s < 2,\ s \ne 0). \tag{A.15}$$

**Lemma A.4** (A lower bound for the real part of the digamma function)**.** *For real $x$ with $\frac{1}{2} \le x \le 1$ and real $y$,*

$$\operatorname{Re}\psi(x+iy) \ge \frac{1}{2}\log(x^2+y^2) - \gamma - \log 2 + \frac{y^2}{2x\,(x^2+y^2)}. \tag{A.16}$$

*Proof.* By (A.13), $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{l\ge 0} g_\psi(l)$ with $g_\psi(\lambda) = y^2/((\lambda+x)((\lambda+x)^2+y^2))$ for $\lambda \ge 0$. With $v = \lambda + x > 0$ one finds

$$\frac{d^2}{dv^2}\,\frac{1}{v\,(v^2+y^2)} = \frac{12v^4 + 6v^2y^2 + 2y^4}{v^3(v^2+y^2)^3} > 0, \tag{A.17}$$

so $g_\psi$ is convex. For a convex function the chord lies above the graph, so $\int_l^{l+1} g_\psi \le \frac{1}{2}(g_\psi(l) + g_\psi(l+1))$ for every integer $l \ge 0$. Summing over $l$, and using $g_\psi(\lambda) \to 0$ as $\lambda \to \infty$, gives $\int_0^\infty g_\psi \le \sum_{l\ge 0} g_\psi(l) - \frac{1}{2}g_\psi(0)$. Since $\int_0^\infty g_\psi = \int_x^\infty y^2\,dv/(v(v^2+y^2)) = \frac{1}{2}\log(1+y^2/x^2)$, we obtain $\operatorname{Re}\psi(x+iy) \ge \psi(x) + \frac{1}{2}\log(1+y^2/x^2) + y^2/(2x(x^2+y^2))$. Termwise differentiation of (A.13) for real $x > 0$ gives $\frac{d}{dx}(\psi(x) - \log x) = \sum_{l\ge 0}(l+x)^{-2} - \frac{1}{x} > 0$, because $(l+x)^{-2} > \int_l^{l+1}(\lambda+x)^{-2}d\lambda$ and the integrals add up to $\frac{1}{x}$. By (A.13), $\psi(\frac{1}{2}) = -\gamma - 2\sum_{l\ge 0}(\frac{1}{2l+1} - \frac{1}{2l+2}) = -\gamma - 2\log 2$. Hence for $x \ge \frac{1}{2}$ we have $\psi(x) \ge \log x + \psi(\frac{1}{2}) + \log 2 = \log x - \gamma - \log 2$, and $\log x + \frac{1}{2}\log(1+y^2/x^2) = \frac{1}{2}\log(x^2+y^2)$ gives (A.16). □

**Theorem A.5** ($A$ does not vanish above height 9)**.** *Let $0 \le \sigma \le 1$ and $|t| \ge 9$, and put*

$$\mathcal{Q}(t) = \log(8\pi) - \frac{1}{2}\log(1+t^2) - \frac{t^2}{2(t^2+4)} + \frac{1}{t^2} - 0.64\,\frac{t^2-1}{t^2+4} + 10^{-27}. \tag{A.18}$$

*Then $\operatorname{Re}(sA(s)) \le \mathcal{Q}(t) < 0$ for $s = \sigma + it$. In particular $A(s) \ne 0$.*

*Proof.* Since $A(\bar{s}) = \overline{A(s)}$ by (4.3), we may assume $t \ge 9$. We bound the real parts of the terms of (A.15). First, $\operatorname{Re}(1/s) = \sigma/(\sigma^2+t^2) \le 1/t^2$. Second, apply Lemma A.4 with $x = 1 - \frac{\sigma}{2} \in [\frac{1}{2}, 1]$ and $y = -\frac{t}{2}$. Then $x^2 + y^2 \ge \frac{1}{4}(1+t^2)$, and $x \le 1$, $x^2 + y^2 \le \frac{1}{4}(4+t^2)$ give $y^2/(2x(x^2+y^2)) \ge t^2/(2(t^2+4))$. Hence

$$-\operatorname{Re}\psi\Big(1 - \frac{s}{2}\Big) - \gamma \le 2\log 2 - \frac{1}{2}\log(1+t^2) - \frac{t^2}{2(t^2+4)}. \tag{A.19}$$

Third, by (A.14), $\mathrm{Re}(sQ_0(s)) = \sum_{k\geq 1} \frac{\Sigma_{2k}-1}{k}\,\mathfrak{r}_k$ with

$$\mathfrak{r}_k = \mathrm{Re}\,\frac{s}{2k-s} = \frac{2k\sigma-\sigma^2-t^2}{(2k-\sigma)^2+t^2}. \tag{A.20}$$

For $0\leq\sigma\leq 1$ we have $2k\sigma-\sigma^2\leq 2k-1$, because the left side increases with $\sigma$ on $[0,1]$. For $k=1$ the numerator is at most $1-t^2<0$ and the denominator is at most $4+t^2$, so $\mathfrak{r}_1\leq -(t^2-1)/(t^2+4)$. If $2\leq k$ and $2k-1<t^2$, then $\mathfrak{r}_k<0$. If $2k-1\geq t^2$, then $k\geq 41$ because $t^2\geq 81$, and $\mathfrak{r}_k\leq (2k-1)/(2k-1)^2 = 1/(2k-1)$, since the denominator is at least $(2k-1)^2$. With $\Sigma_{2k}-1\leq 3\cdot 4^{-k}$ these terms add up to at most $\frac{3}{41\cdot 81}\sum_{k\geq 41} 4^{-k} = 4^{-40}/3321 < 10^{-27}$. Finally $\Sigma_2 - 1 > \sum_{m=2}^{10} m^{-2} + \int_{11}^{\infty}\lambda^{-2}d\lambda > 0.5497+0.0909 > 0.64$. Adding up, $\mathrm{Re}(sA(s))\leq \mathcal{Q}(t)$.

Each of the terms $-\frac{1}{2}\log(1+t^2)$, $-\frac{t^2}{2(t^2+4)} = -\frac{1}{2}+\frac{2}{t^2+4}$, $\frac{1}{t^2}$ and $-0.64\frac{t^2-1}{t^2+4} = -0.64+\frac{3.2}{t^2+4}$ decreases for $t>0$, so $\mathcal{Q}(t)\leq\mathcal{Q}(9)$ for $t\geq 9$. Since $e^{3.2242}>8\pi$ and $e^{4.4066}<82$, we have $\log(8\pi)<3.2242$ and $\frac{1}{2}\log 82 > 2.2033$. For these bounds one can use $\pi<355/113$, the lower Taylor sum through degree 20 for $e^{3.2242}$, and the upper bound for $e^{4.4066}$ obtained by adding $\frac{5}{4}(4.4066)^{31}/31!$ to the sum through degree 30. The last bound follows by bounding consecutive term ratios in the tail by $1/5$. With $\frac{81}{170}>0.4764$, $\frac{1}{81}<0.0124$ and $0.64\cdot\frac{80}{85}>0.6023$ this gives $\mathcal{Q}(9) < 3.2242-2.2033-0.4764+0.0124-0.6023+10^{-27} < -0.045$. □

For the bounded rectangle, the program `A_nonvanishing_check.py` uses the holomorphic function $s^2A(s)$, which has value 1 at 0 by (4.3). For $c_k=\Sigma_{2k}-1$, it encloses $c_k$ by the rational sum $\sum_{m=2}^{40} m^{-2k}$ with a nonnegative error at most $40^{1-2k}/(2k-1)$ by the integral test. On $\mathrm{Re}\,s\leq 1$ the part of (A.14) with $k>80$ has modulus at most

$$\frac{\sum_{k>80} 3\cdot 4^{-k}}{81\cdot 161} = \frac{4^{-80}}{81\cdot 161},$$

using $|2k-s|\geq 2k-1$. Multiplying (A.15) by $s$ gives the expression enclosed by the program on each box. The initial boxes tile $0\leq \mathrm{Re}\,s\leq 1$, $0\leq \mathrm{Im}\,s\leq 102$. Subdivision replaces each undecided box by four boxes covering it. A box is accepted when at least one of the real and imaginary enclosures excludes zero. The recorded run accepted 188,892 boxes at a maximum subdivision depth of 5, with no undecided boxes. Its smallest recorded lower bound for $\max\{|\mathrm{Re}(s^2A(s))|, |\mathrm{Im}(s^2A(s))|\}$ was enclosed by $[1.00128\cdot 10^{-5} \pm 3.76\cdot 10^{-11}]$. Thus $A$ is nonzero on the certified rectangle where $s\neq 0$. Together with the analytic estimate and conjugation symmetry, this gives nonvanishing on $0\leq \mathrm{Re}\,s\leq 1$ wherever $A$ is holomorphic.

## Appendix B. Figures

The surface and zero-map panels are numerical illustrations. The six discs in Appendix A identify which computed zeros of $G$ have enclosures. The five modulus surfaces and the zero map were regenerated for this version. Each red marker on a surface uses the computed modulus at the corresponding mesh vertex. On the real interval $s > -2$, (A.3) gives $H(s) > 0$, since $\mathcal{K}$ is positive off the integers. Equation (A.4) therefore gives the sign of $G$ shown in the real-axis plot and a simple zero at $-1$.

For the period mean in the Gram plot, use the absolutely convergent cosine series for $\log|1-\alpha e^{2\pi i u}|$ from the proof of (A.4). The logarithmic majorant used there is square integrable, so these functions converge to $h$ in $L^2$ on a period. Their products at $x/j$ and $x/k$ converge in $L^1$ on a common period by Cauchy–Schwarz. Only equal frequencies $m/j = n/k$ survive in the period mean, with cosine product mean $1/2$. Writing $(m,n) = r(j/g_{jk}, k/g_{jk})$ with $r \geq 1$ and $g_{jk} = \gcd(j,k)$, their contribution is

$$\frac{1}{2}\sum_{r\geq 1}\frac{g_{jk}^2}{r^2 jk} = \frac{\pi^2}{12}\frac{g_{jk}^2}{jk}.$$

For the weighted panel the same $L^1$ convergence holds after summing over periods against the weight $x^{-2\sigma-1}$ on $[\max(j,k),\infty)$, since $\sigma > 0$. Upon writing $x = \max(j,k)v$, the cosine product formula replaces each pair of modes by cosines with arguments $2\pi\max(j,k)(m/j \pm n/k)v$. The kernel applied to each frequency $\beta$ is $2\sigma\int_1^\infty v^{-2\sigma-1}\cos(\beta v)\,dv$. It equals 1 at zero and has modulus at most $\min(1, 4\sigma/|\beta|)$ otherwise, by integration by parts. At fixed $m$, the difference-frequency terms are bounded in sum by $O((1+\log m)/m^2)$: below $mk/(2j)$ the denominator $|mk-nj|$ is $\gg m$, above $2mk/j$ it is $\gg n$, and between them $n$ is comparable to $m$ and the nonzero $mk-nj$ are spaced by $j$. The equal-frequency terms contribute $O(m^{-2})$, and the sum-frequency terms are bounded by the same estimate using $mk+nj \gg m+n$. This proves absolute convergence, so the frequency groups can be summed before truncating for the plot.

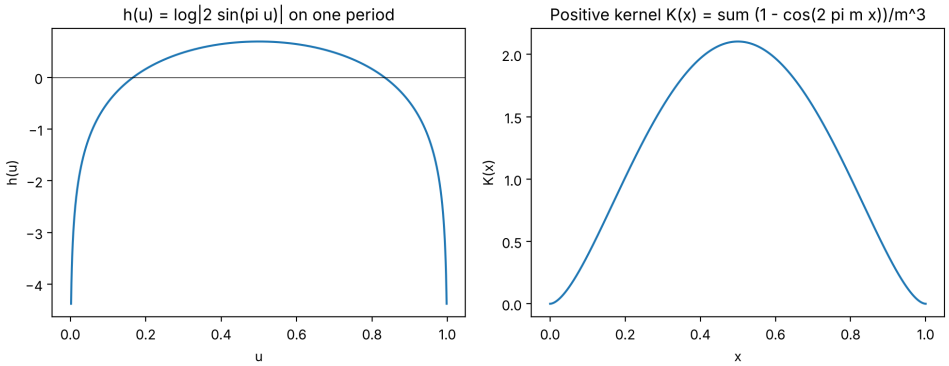



FIGURE 1. The function $h$ on one period and the positive kernel $\mathcal{K}(u) = \sum_{m\geq 1}(1-\cos(2\pi m u))/m^3$.

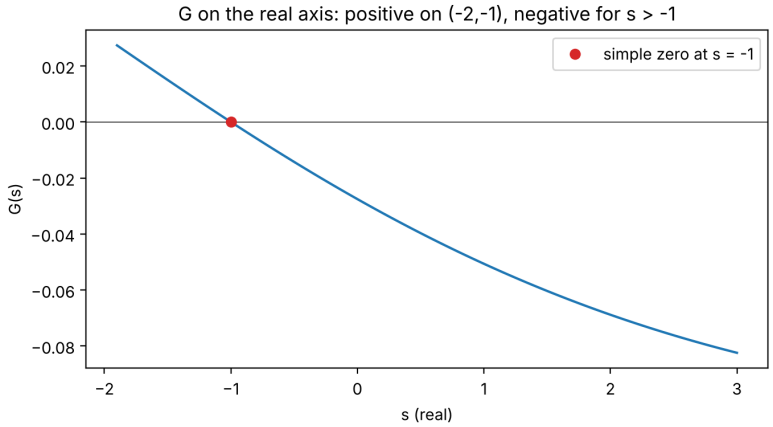



FIGURE 2. $G$ on the real axis.

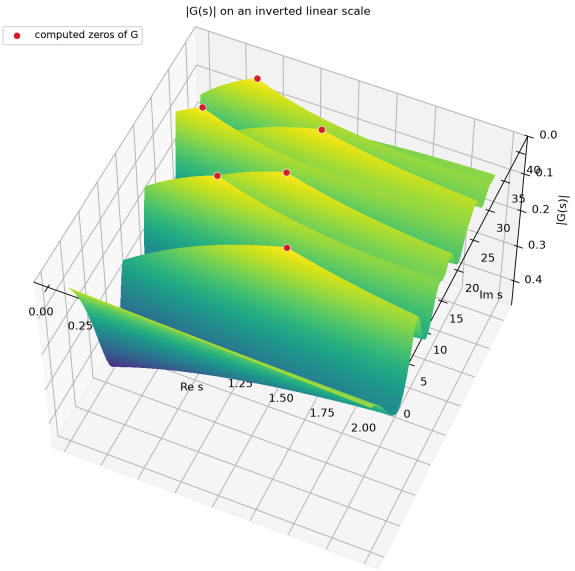



FIGURE 3. $|G(s)|$ on a linear scale with the vertical axis inverted. The red markers indicate the six computed zeros of $G$ in the certified discs.

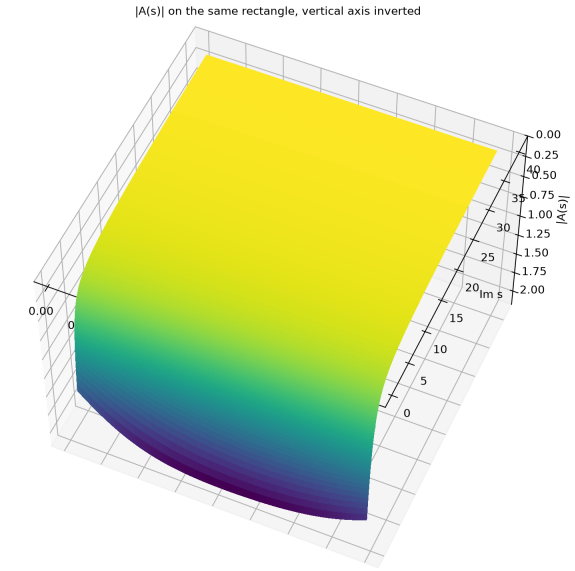



FIGURE 4. $|A(s)|$ on the same rectangle, with the vertical axis inverted. The nonvanishing result of Appendix A covers the part with $0 \le \operatorname{Re} s \le 1$.

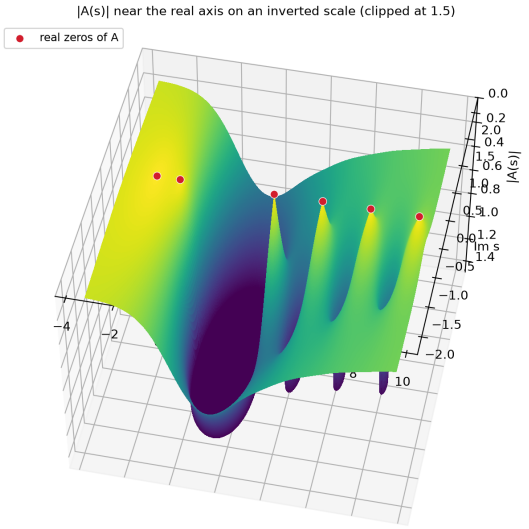



FIGURE 5. $|A(s)|$ near the real axis, clipped above 1.5. The red markers indicate computed real zeros of $A$.

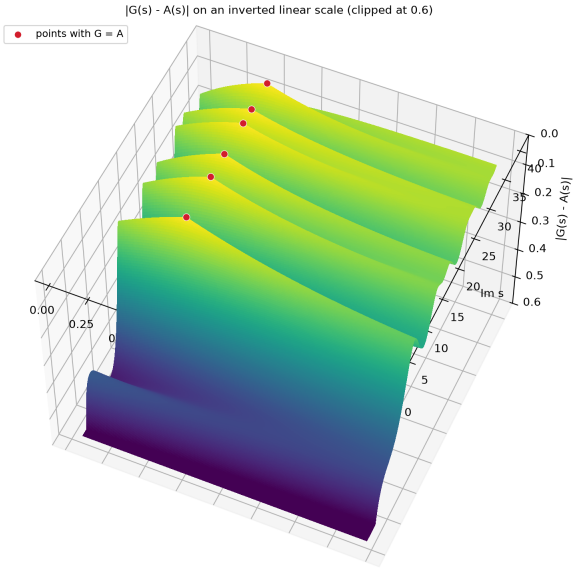



FIGURE 6. $|G(s) - A(s)|$, clipped above 0.6, with the vertical axis inverted.

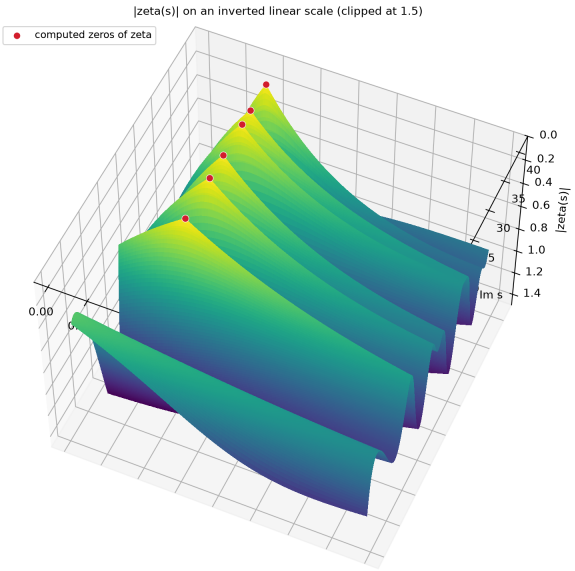



FIGURE 7. $|\zeta(s)|$, clipped above 1.5, with the vertical axis inverted. The red markers indicate the first six nontrivial zeros in the upper half-plane.

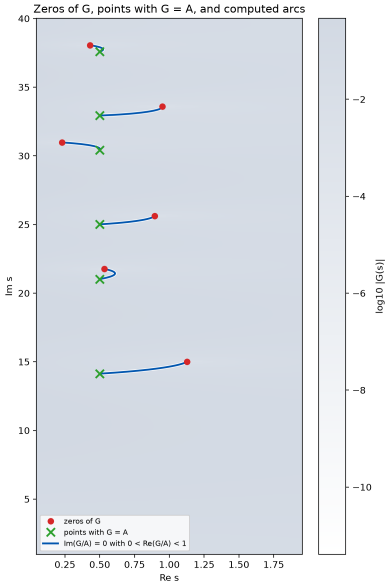



FIGURE 8. Computed zeros of $G$, points with $G = A$, and arcs of $\mathrm{Im}(G/A) = 0$ with $0 < \mathrm{Re}(G/A) < 1$ where $A \neq 0$. The pairing shown by the arcs is a numerical observation.

Let $z_b^G$ be the computed zeros of $G$ used in the plots and $z_b^A$ the computed real zeros of $A$. Their counts are $n_G = 79$ and $n_A = 101$. The exponential sums are

$$\mathcal{E}_G(x) = \frac{1}{n_G}\sum_{b=1}^{n_G} e^{ix\,\mathrm{Im}\, z_b^G}, \qquad \mathcal{E}_A(x) = \frac{1}{n_A}\sum_{b=1}^{n_A} e^{ixz_b^A}. \tag{B.1}$$

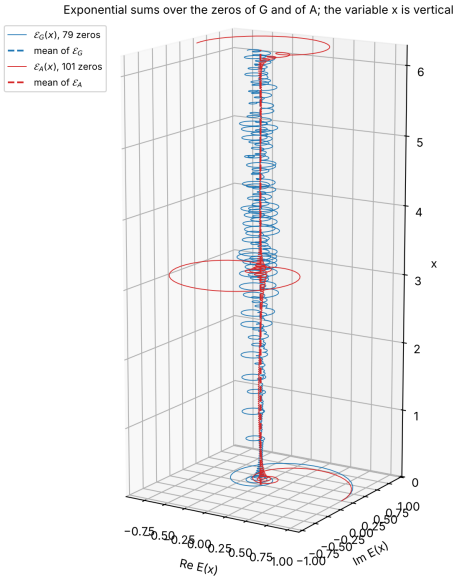



FIGURE 9. The exponential sums $\mathcal{E}_G(x)$ and $\mathcal{E}_A(x)$ for $0 \le x \le 2\pi$. The variable $x$ is the vertical axis.

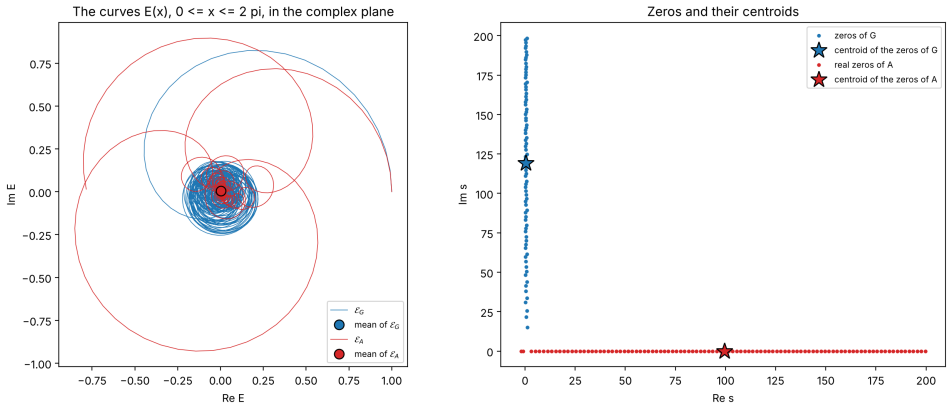



FIGURE 10. The two exponential sums in the complex plane and the computed zeros used in the sums, with their centroids.

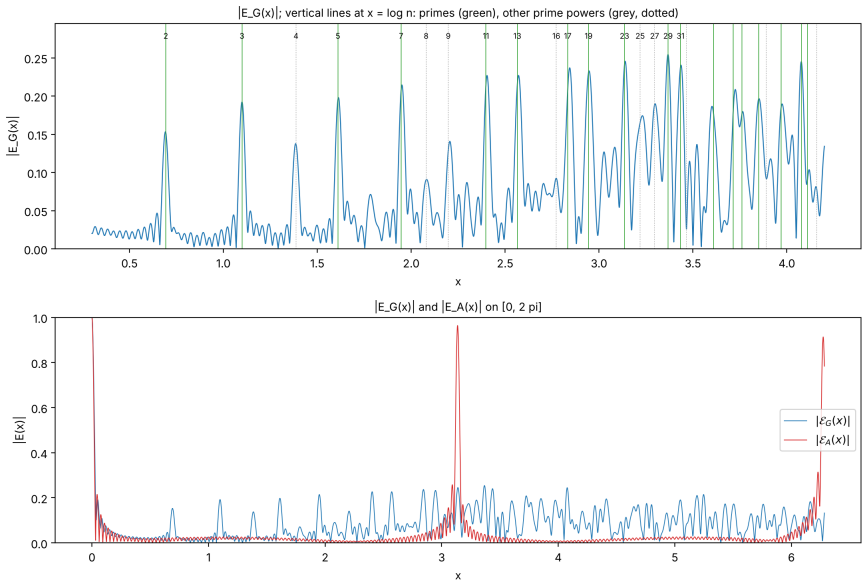



FIGURE 11. Moduli of the exponential sums. The vertical lines in the upper panel are at $x = \log n$ for prime powers $n \leq 66$.

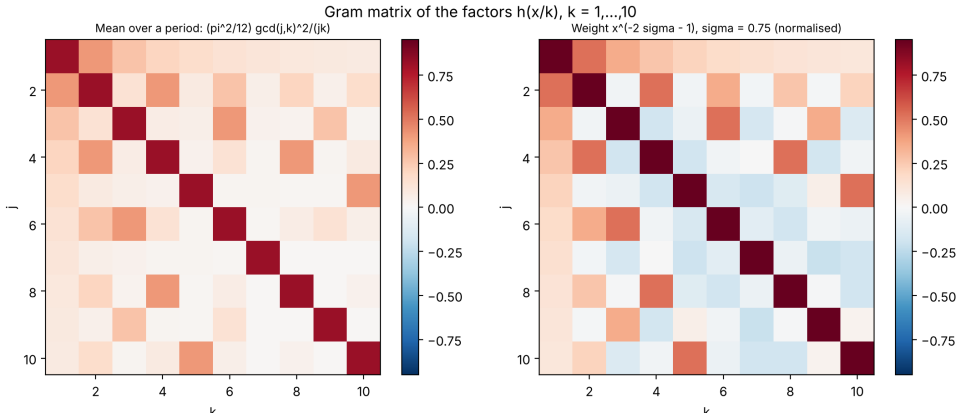



FIGURE 12. Gram matrices of $h(x/k)$ for $1 \leq k \leq 10$. The first uses the mean over a common period. For the second, the integral with weight $x^{-2\sigma-1}$ starts at $\max(j,k)$ and is normalized by $2\sigma \max(j,k)^{2\sigma}$, with $\sigma = 3/4$.

## Appendix C. The vertical cut and the Python sources

The interactive vertical cut in `G_A_zeta_cut_v2.html` shows $t \mapsto G(\frac{1}{2}+it)$, $t \mapsto A(\frac{1}{2}+it)$ and their difference. On the sampled range $0.5 \le t \le 100$ with step 0.01, the accompanying output reports 30 local minima, of which 29 met the numerical threshold $|G-A| \le 10^{-6}$ after refinement. The ordinates of these 29 points agree with the corresponding computed zeta ordinates within $6 \cdot 10^{-7}$. The threshold and agreement are finite numerical observations. The HTML, the script, and its output accompany this paper.

The following Python sources generate the static figures and the data for the vertical cut. The accompanying directory supplies `cp_eval.py` and the zero lists used by the plotting programs. Run the four static plotting scripts in the order printed below. The first generates the kernel and real-axis plots. The vertical-cut listing uses a relative path to the evaluator.

Listing 1. `figures_kernel_real.py`

```
# Kernel and real-axis plots. G is computed from the positive kernel K:
#   G(s) = -(s+1)(s+2) H(s)/(4 pi^2),  H(s) = int_0^1 K(x) sum_{n>=1} (n+x)^{-s-3} dx
     (midpoint rule + Euler-Maclaurin tail).
import numpy as np
import matplotlib; matplotlib.use('Agg')
import matplotlib.pyplot as plt
M, N, L = 2000, 24, 8192
x = (np.arange(L) + 0.5)/L
K = np.zeros(L)
for m0 in range(1, M+1, 200):
    m = np.arange(m0, min(m0+200, M+1))[:, None]
    K += ((1 - np.cos(2*np.pi*m*x[None, :]))/m**3).sum(axis=0)
def H(s):
    a = s + 3; tot = 0j
    for n in range(1, N+1): tot += (K*(n + x)**(-a)).sum()
    b = N + 1 + x
    tot += (K*(b**(1-a)/(a-1) + b**(-a)/2 + a*b**(-a-1)/12 - a*(a+1)*(a+2)*b**(-a-3)
     /720)).sum()
    return tot/L
def G(s): return -(s+1)*(s+2)*H(s)/(4*np.pi**2)
# kernel
u = np.linspace(0.002, 0.998, 2000)
fig, axs = plt.subplots(1, 2, figsize=(10, 4))
axs[0].plot(u, np.log(2*np.sin(np.pi*u))); axs[0].axhline(0, color='k', lw=0.5); axs
     [0].set_title('h(u) = log|2 sin(pi u)| on one period'); axs[0].set_xlabel('u');
     axs[0].set_ylabel('h(u)')
axs[1].plot(x, K); axs[1].set_title('Positive kernel K(x) = sum (1 - cos(2 pi m x))/m
     ^3'); axs[1].set_xlabel('x'); axs[1].set_ylabel('K(x)')
fig.tight_layout(); fig.savefig('fig_kernel.png', dpi=200)
# G on the real axis
sr = np.linspace(-1.9, 3, 99); Gr = np.array([G(complex(v, 0)).real for v in sr])
fig, ax = plt.subplots(figsize=(7, 4)); ax.plot(sr, Gr); ax.axhline(0, color='k', lw
     =0.5); ax.plot([-1], [0], 'o', color='tab:red', label='simple zero at s = -1')
ax.set_xlabel('s (real)'); ax.set_ylabel('G(s)'); ax.set_title('G on the real axis:
     positive on (-2,-1), negative for s > -1'); ax.legend()
fig.tight_layout(); fig.savefig('fig_G_real.png', dpi=200)
print("G(1) =", G(1+0j).real, " G(-1) =", G(-1+0j).real)
```

Listing 2. figure_gram.py

```
# Castor-Pollux I: heat maps of the Gram matrix of the factors h(x/k), k <= 10 (labels in English).
# Left: mean over a period, (pi^2/12) gcd(j,k)^2/(jk).  Right: weight x^{-2 sigma-1}, sigma = 3/4, normalised bracket
#   2 sigma max(j,k)^{2 sigma} I_{jk}(sigma) = T(0) + sum_r D(r) kappa(2 pi max(j,k) r/(jk))   (series truncated at r <= R).
import mpmath as mp, numpy as np
from math import gcd
import matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
mp.mp.dps = 15
def T(j,k,r):
    g = gcd(j,k); jp, kp = j//g, k//g
    if r % g: return mp.mpf(0)
    rp = r//g
    if rp == 0: return mp.pi**2/(12*jp*kp)
    m = 1
    while (m*kp - rp) % jp or (m*kp - rp)//jp < 1: m += 1
    n = (m*kp - rp)//jp
    return (mp.digamma(mp.mpf(m)/jp) - mp.digamma(mp.mpf(n)/kp))/(2*rp)
def U(j,k,r):
    s = mp.mpf(0)
    for m in range(1, r//k + 1):
        if (r - m*k) % j == 0 and (r - m*k)//j >= 1: s += mp.mpf(1)/(2*m*((r - m*k)//j))
    return s
def kappa(sig, y): return 2*sig*mp.re(mp.expint(2*sig+1, -1j*y))
K, sig, R = 10, mp.mpf('0.75'), 1500
flat = np.zeros((K,K)); wt = np.zeros((K,K))
for j in range(1, K+1):
    for k in range(j, K+1):
        flat[j-1,k-1] = flat[k-1,j-1] = float(T(j,k,0))
        tot = T(j,k,0); g = gcd(j,k)
        for r in range(g, R+1, g): tot += (T(j,k,r) + T(j,k,-r) + U(j,k,r))*kappa(sig, 2*mp.pi*max(j,k)*r/(j*k))
        wt[j-1,k-1] = wt[k-1,j-1] = float(tot)
np.savetxt('gram_flat.csv', flat, delimiter=',', fmt='%.8f'); np.savetxt('gram_weighted.csv', wt, delimiter=',', fmt='%.8f')
print("eigenvalues of the mean Gram matrix:", np.round(np.linalg.eigvalsh(flat), 5))
print("eigenvalues of the normalised weighted bracket matrix:", np.round(np.linalg.eigvalsh(wt), 5))
print("negative entries of the weighted bracket:", int((wt < 0).sum()), "of", K*K)
fig, axs = plt.subplots(1, 2, figsize=(11, 4.6)); v = max(abs(flat).max(), abs(wt).max())
for ax, Mx, title in ((axs[0], flat, 'Mean over a period: (pi^2/12) gcd(j,k)^2/(jk)'), (axs[1], wt, 'Weight x^(-2 sigma - 1), sigma = 0.75 (normalised)')):
    im = ax.imshow(Mx, cmap='RdBu_r', vmin=-v, vmax=v, extent=(0.5, K+0.5, K+0.5, 0.5)); ax.set_xlabel('k'); ax.set_ylabel('j'); ax.set_title(title, fontsize=9); fig.colorbar(im, ax=ax)
fig.suptitle('Gram matrix of the factors h(x/k), k = 1,...,10'); fig.tight_layout(); fig.savefig('fig_gram.png', dpi=200)
```

Listing 3. figures_v4.py

```
# Surfaces on a locally refined grid, with linear inverted vertical axes.
# Each red marker uses the computed modulus at its mesh vertex.
```

```
# G and A are evaluated with cp_eval.py (kernel K only; no zeta function).  zeta is used only for its own figure,
# for the six points 1/2 + i t and for the check of the grid relation.
import time
import numpy as np
import mpmath as mp
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from matplotlib.colors import LinearSegmentedColormap
from matplotlib.patheffects import Normal, Stroke
import cp_eval as E

out = open('figures_v4_output.txt', 'w')


def log(s=''):
    print(s, flush=True)
    out.write(s + '\n')


def zeta_grid(sig, t):
    try:
        from flint import acb, ctx
        ctx.prec = 64
        f = lambda a, b: complex(acb(a, b).zeta())
    except ImportError:
        f = lambda a, b: complex(mp.zeta(mp.mpc(a, b)))
    return np.array([[f(float(a), float(b)) for a in sig] for b in t])


# ------------------------------------------------------------------ data
zg = np.loadtxt('zeros_G.txt')
zG = zg[:, 0] + 1j * zg[:, 1]
zG40 = zG[zG.imag <= 40]
mp.mp.dps = 20
zD = np.array([0.5 + 1j * float(mp.zetazero(k).imag) for k in range(1, 7)])      # points 1/2 + i t
log('zeros of G with Im s <= 40: %d;  points 1/2 + i t: %d' % (len(zG40), len(zD)))

# local refinement: grid lines through every zero and at these distances from it, so that each peak is resolved
OFF_S = np.array([-0.018, -0.009, -0.003, -0.001, 0.0, 0.001, 0.003, 0.009, 0.018])
OFF_T = 12 * OFF_S
pts = np.concatenate([zG40, zD])
sig = np.concatenate([np.round(np.arange(0.05, 1.9501, 0.01), 2)] + [z.real + OFF_S for z in pts])
sig = np.unique(sig[(sig >= 0.05) & (sig <= 1.95)])
t_grid = np.concatenate([np.round(np.arange(1.0, 40.0001, 0.05), 2)] + [z.imag + OFF_T for z in pts])
tt = np.unique(t_grid[(t_grid >= 1.0) & (t_grid <= 40.0)])
S, T = np.meshgrid(sig, tt)
t0 = time.time()
Gv = E.G_grid(sig, tt)
Av = E.A(S + 1j * T)
Zv = zeta_grid(sig, tt)
```

```
log('fine grid: %d (Re s) x %d (Im s) points, %.0f s' % (len(sig), len(tt), time.time() - t0))
sc = S + 1j * T
chk = np.abs(Gv - Av - (np.pi / sc) / np.tan(np.pi * sc / 2) * Zv).max()
log('max |G - A - (pi/s) cot(pi s/2) zeta| on the fine grid = %.3e' % chk)
k = np.unravel_index(np.abs(Av).argmin(), Av.shape)
log('min |A| on the fine grid = %.6f at s = %.2f + %.2f i;  max |A| = %.4f' % (np.abs(Av).min(), sig[k[1]], tt[k[0]], np.abs(Av).max()))
gd = np.abs(E.G(zD) - E.A(zD))
log('|G - A| at the six points 1/2 + i t: ' + ' '.join('%.1e' % v for v in gd))
log('|G| at the six zeros of G (double precision evaluator): ' + ' '.join('%.1e' % v for v in np.abs(E.G(zG40))))
np.savez_compressed('grid_values_v4.npz', sigma=sig, t=tt, G=Gv, A=Av, zeta=Zv)


def surface(V, title, fname, zlabel, marks, marklabel, Sg=S, Tg=T, elev=52, azim=-66, top=None, figsize=(9, 7.5)):
    """|F| on a linear vertical axis, inverted (0 at the top): a zero of F is a peak whose tip is at 0."""
    Zs = np.abs(V)
    if top is not None:
        Zs = np.minimum(Zs, top)
    columns = np.searchsorted(Sg[0], marks.real)
    rows = np.searchsorted(Tg[:, 0], marks.imag)
    if not (np.array_equal(Sg[rows, columns], marks.real)
            and np.array_equal(Tg[rows, columns], marks.imag)):
        raise ValueError(f'{fname}: each marked zero must be a mesh vertex')
    fig = plt.figure(figsize=figsize)
    ax = fig.add_subplot(111, projection='3d', computed_zorder=False)
    ax.plot_surface(Sg, Tg, Zs, cmap='viridis_r', zorder=1, linewidth=0, antialiased=False, rcount=Sg.shape[0], ccount=Sg.shape[1])
    if len(marks):
        ax.scatter(marks.real, marks.imag, Zs[rows, columns],
                   color='#d21d2c', edgecolor='white', linewidth=0.5,
                   s=40, depthshade=False, label=marklabel, zorder=10)
        ax.legend(loc='upper left', fontsize=9)
    ax.set_xlabel('Re s'); ax.set_ylabel('Im s'); ax.set_zlabel(zlabel); ax.set_title(title, fontsize=10)
    ax.set_zlim(0, Zs.max()); ax.invert_zaxis()
    ax.view_init(elev=elev, azim=azim)
    fig.tight_layout(); fig.savefig(fname, dpi=200); plt.close(fig)
    if len(marks):
        log('%s: %d marked mesh vertices, largest plotted marker height %.3e'
            % (fname, len(marks), Zs[rows, columns].max()))
    else:
        log('wrote %s' % fname)


none = np.array([], complex)
surface(Gv, '|G(s)| on an inverted linear scale', 'fig_G_surface.png', '|G(s)|', zG40, 'computed zeros of G')
surface(Av, '|A(s)| on the same rectangle, vertical axis inverted', 'fig_A_surface.png', '|A(s)|', none, '')
surface(Gv - Av, '|G(s) - A(s)| on an inverted linear scale (clipped at 0.6)', 'fig_GminusA_surface.png', '|G(s) - A(s)|', zD, 'points with G = A', top=0.6)
surface(Zv, '|zeta(s)| on an inverted linear scale (clipped at 1.5)', 'fig_zeta_surface.png', '|zeta(s)|', zD, 'computed zeros of zeta', top=1.5)
```

```
100
101 # ------------------------------------------------------------- A near the real axis
        : real zeros and poles
102 zA_all = np.loadtxt('zeros_A.txt')[:, 0]
103 zA = zA_all[(zA_all > -3.5) & (zA_all < 9.5)]
104 OFF_A = np.array([-0.03, -0.015, -0.006, -0.002, 0.0, 0.002, 0.006, 0.015, 0.03])
105 sa = np.unique(np.concatenate([np.arange(-3.495, 9.5, 0.01)] + [z + OFF_A for z in zA
        ]))
106 ta = np.unique(np.concatenate([np.round(np.arange(-2.0, 2.0001, 0.01), 2), OFF_A]))
107 Sa, Ta = np.meshgrid(sa, ta)
108 Aa = E.A(Sa + 1j * Ta)
109 log('A near the real axis: %d x %d points; real zeros shown: %s' % (len(sa), len(ta),
        ' '.join('%.4f' % v for v in zA)))
110 surface(Aa, '|A(s)| near the real axis on an inverted scale (clipped at 1.5)',
111         'fig_A_real_surface.png', '|A(s)|', zA.astype(complex), 'real zeros of A', Sg=
        Sa, Tg=Ta, elev=48, azim=-78, top=1.5, figsize=(11, 6.5))
112
113 # ------------------------------------------------------------- zero map on the fine
         grid
114 Psi = Gv / Av
115 fig, ax = plt.subplots(figsize=(6, 9))
116 background = LinearSegmentedColormap.from_list('light_gray', ['#ffffff', '#d2d9e3'])
117 log_modulus = np.log10(np.maximum(np.abs(Gv), np.finfo(float).tiny))
118 pc = ax.pcolormesh(S, T, log_modulus, cmap=background, shading='auto')
119 fig.colorbar(pc, ax=ax, label='log10 |G(s)|')
120 arc_color = '#0055ae'
121 arcs = ax.contour(S, T, np.where((Psi.real > 0) & (Psi.real < 1), Psi.imag, np.nan),
122                   levels=[0], colors=[arc_color], linewidths=2.0)
123 arcs.set_path_effects([Stroke(linewidth=3.6, foreground='white'), Normal()])
124 ax.plot(zG40.real, zG40.imag, 'o', color='tab:red', label='zeros of G')
125 ax.plot(zD.real, zD.imag, 'x', color='tab:green', ms=9, mew=2, label='points with G =
        A')
126 ax.plot([], [], color=arc_color, linewidth=2.0, label='Im(G/A) = 0 with 0 < Re(G/A) <
        1')
127 ax.set_xlabel('Re s'); ax.set_ylabel('Im s'); ax.set_title('Zeros of G, points with G
        = A, and computed arcs'); ax.legend(loc='lower left', fontsize=8)
128 fig.tight_layout(); fig.savefig('fig_zero_map.png', dpi=200); plt.close(fig)
129 log('wrote fig_zero_map.png')
130 out.close()
```

## Listing 4. expsum_v3.py

```
1 # Castor-Pollux I v3: exponential sums over the computed zeros of G and of A (all
        labels in English).
2 #   E_G(x) = (1/79)  sum_b exp(i x Im z_b),  z_b the zeros of G with 0.2 <= Im s <= 200
          (zeros_G.txt),
3 #   E_A(x) = (1/101) sum_b exp(i x z_b),     z_b the real zeros of A in [-60, 200]
        (zeros_A.txt),
4 # for 0 <= x <= 2 pi.  In the 3D figure the variable x is the vertical axis.
5 # The zero lists were computed in the branch castor-pollux-zeros of vcasana1/PaperMath
        and are taken as given.
6 import math
7 import numpy as np
8 import matplotlib
9 matplotlib.use('Agg')
10 import matplotlib.pyplot as plt
11
```

```
XMAX = 2 * np.pi
X = np.linspace(0, XMAX, 6001)
zg = np.loadtxt('zeros_G.txt'); zG = zg[:, 0] + 1j * zg[:, 1]
zA = np.loadtxt('zeros_A.txt')[:, 0]
freq = {'G': zG.imag, 'A': zA}
Ev = {n: np.exp(1j * np.outer(X, f)).sum(1) / len(f) for n, f in freq.items()}
col = {'G': 'tab:blue', 'A': 'tab:red'}
out = open('expsum_v3_output.txt', 'w')


def log(s=''):
    print(s); out.write(s + '\n')


cen = {}
for n, f in freq.items():
    c = ((np.exp(1j * f * XMAX) - 1) / (1j * f * XMAX)).mean()       # (1/(2 pi))
     int_0^{2 pi} E(x) dx, closed form
    cen[n] = c
    log('%s: %d zeros, frequencies in [%.4f, %.4f]' % (n, len(f), f.min(), f.max()))
    log('   mean of the curve E_%s on [0, 2 pi] = %.5f %+.5f i   (trapezoid rule: %.5f
      %+.5f i)'
       % (n, c.real, c.imag, (np.trapezoid(Ev[n], X) / XMAX).real, (np.trapezoid(Ev[n
     ], X) / XMAX).imag))
    log('   mean of |E_%s(x)| on [0, 2 pi] = %.4f,  max over x >= 0.3: %.4f' % (n, np.
     abs(Ev[n]).mean(), np.abs(Ev[n])[X >= 0.3].max()))
log('centroid of the zeros of G: %.5f %+.5f i;  Re s in [%.4f, %.4f]' % (zG.mean().
     real, zG.mean().imag, zG.real.min(), zG.real.max()))
log('centroid of the real zeros of A: %.5f' % zA.mean())


def is_prime(n): return n > 1 and all(n % d for d in range(2, int(n ** 0.5) + 1))
def is_prime_power(n):
    for p in range(2, n + 1):
        if is_prime(p) and n % p == 0:
            while n % p == 0: n //= p
            return n == 1
    return False


log('')
log('E_G(log n) for 2 <= n <= 31:   n   kind   Re   Im   modulus')
vals = {}
for n in range(2, 32):
    v = np.exp(1j * freq['G'] * math.log(n)).mean()
    kind = 'prime' if is_prime(n) else ('prime power' if is_prime_power(n) else 'other
     ')
    vals[n] = (kind, v)
    log('%3d  %-11s  %+.3f  %+.3f  %.3f' % (n, kind, v.real, v.imag, abs(v)))
for kind in ('prime', 'prime power', 'other'):
    m = [abs(v) for k, v in vals.values() if k == kind]
    r = [v.real for k, v in vals.values() if k == kind]
    log('%-11s: modulus between %.3f and %.3f; real part between %+.3f and %+.3f' % (
     kind, min(m), max(m), min(r), max(r)))
xa = X[(X > 0.3)]
log('E_A: |E_A(pi)| = %.4f, |E_A(2 pi)| = %.4f; max of |E_A| on [0.3, 2 pi] outside |x
      - pi| < 0.2 and x > 2 pi - 0.2: %.4f'
```

```
    % (abs(np.exp(1j * zA * np.pi).mean()), abs(np.exp(1j * zA * 2 * np.pi).mean()),
       np.abs(Ev['A'])[(X > 0.3) & (np.abs(X - np.pi) > 0.2) & (X < 2 * np.pi - 0.2)].
    max()))
out.close()

# ---- 3D: x is the vertical axis
fig = plt.figure(figsize=(8, 11))
ax = fig.add_subplot(111, projection='3d')
for n in ('G', 'A'):
    ax.plot(Ev[n].real, Ev[n].imag, X, color=col[n], lw=0.7, label=r'$\mathcal{E}_%s(x
    )$, %d zeros' % (n, len(freq[n])))
    ax.plot([cen[n].real] * 2, [cen[n].imag] * 2, [0, XMAX], color=col[n], ls='--', lw
    =1.5, label=r'mean of $\mathcal{E}_%s$' % n)
ax.set_xlabel('Re E(x)'); ax.set_ylabel('Im E(x)'); ax.set_zlabel('x')
ax.set_zlim(0, XMAX); ax.set_box_aspect((1, 1, 2.2))
ax.set_title('Exponential sums over the zeros of G and of A; the variable x is
    vertical', fontsize=10)
ax.legend(loc='upper left', fontsize=8); ax.view_init(elev=14, azim=-58)
fig.savefig('fig_expsum_3d.png', dpi=200, bbox_inches='tight', pad_inches=0.15); plt.
    close(fig)

# ---- the curves in the complex plane, and the zeros with their centroids
fig2, ax2 = plt.subplots(1, 2, figsize=(13, 5.5))
for n in ('G', 'A'):
    ax2[0].plot(Ev[n].real, Ev[n].imag, color=col[n], lw=0.6, label=r'$\mathcal{E}_%s$
    ' % n)
    ax2[0].plot(cen[n].real, cen[n].imag, 'o', color=col[n], ms=9, mec='k', label=r'
    mean of $\mathcal{E}_%s$' % n)
ax2[1].plot(zG.real, zG.imag, '.', color=col['G'], ms=5, label='zeros of G')
ax2[1].plot(zG.mean().real, zG.mean().imag, '*', color=col['G'], ms=16, mec='k', label
    ='centroid of the zeros of G')
ax2[1].plot(zA, 0 * zA, '.', color=col['A'], ms=5, label='real zeros of A')
ax2[1].plot(zA.mean(), 0, '*', color=col['A'], ms=16, mec='k', label='centroid of the
    zeros of A')
ax2[0].set_xlabel('Re E'); ax2[0].set_ylabel('Im E'); ax2[0].set_aspect('equal')
ax2[0].set_title('The curves E(x), 0 <= x <= 2 pi, in the complex plane'); ax2[0].
    legend(fontsize=8)
ax2[1].set_xlabel('Re s'); ax2[1].set_ylabel('Im s'); ax2[1].set_title('Zeros and
    their centroids'); ax2[1].legend(fontsize=8)
fig2.tight_layout(); fig2.savefig('fig_expsum_plane.png', dpi=200); plt.close(fig2)

# ---- moduli
fig3, ax3 = plt.subplots(2, 1, figsize=(11, 7.5))
xx = np.linspace(0.3, 4.2, 20001)
mg = np.abs(np.exp(1j * np.outer(xx, freq['G'])).mean(1))
ax3[0].plot(xx, mg, color=col['G'], lw=0.9)
for n in range(2, 67):
    if is_prime_power(n):
        ax3[0].axvline(math.log(n), color='0.6' if not is_prime(n) else 'tab:green',
    lw=0.6, ls=':' if not is_prime(n) else '-')
        if n <= 31: ax3[0].text(math.log(n), 0.275, str(n), ha='center', fontsize=7)
ax3[0].set_ylim(0, 0.295); ax3[0].set_xlabel('x'); ax3[0].set_ylabel('|E_G(x)|')
ax3[0].set_title('|E_G(x)|; vertical lines at x = log n: primes (green), other prime
    powers (grey, dotted)', fontsize=10)
for n in ('G', 'A'):
    ax3[1].plot(X, np.abs(Ev[n]), color=col[n], lw=0.7, label=r'$|\mathcal{E}_%s(x)|$'
     % n)
```

```
ax3[1].set_xlabel('x'); ax3[1].set_ylabel('|E(x)|'); ax3[1].set_ylim(0, 1); ax3[1].legend()
ax3[1].set_title('|E_G(x)| and |E_A(x)| on [0, 2 pi]', fontsize=10)
fig3.tight_layout(); fig3.savefig('fig_expsum_modulus.png', dpi=200); plt.close(fig3)
```

## Listing 5. vertical_cut_portable.py

```
#!/usr/bin/env python3
# Castor and Pollux I: vertical cut Re s = 1/2. G and A (from their definitions, no zeta) on s = 1/2 + it, 0.5 <= t <= 100;
# the values of t where G(1/2+it) = A(1/2+it), located from G and A only, and compared afterwards with the zeros of zeta.
import sys, json, time
from pathlib import Path
sys.path.insert(0, str(Path(__file__).resolve().parent.parent))
import numpy as np, mpmath as mp
from scipy.optimize import minimize_scalar
from cp_eval import G_grid, A
t0 = time.time(); out = open('vertical_cut_output.txt', 'w')
def log(s): print(s, flush=True); out.write(s + '\n')
T = np.round(np.arange(0.5, 100.0001, 0.01), 2)
g = G_grid([0.5], T)[:, 0]; a = A(0.5 + 1j * T); d = np.abs(g - a)
cand = [i for i in range(1, len(T) - 1) if d[i] <= d[i - 1] and d[i] <= d[i + 1]]
log('local minima of |G-A| on the grid (step 0.01): %d' % len(cand))
f = lambda t: abs(G_grid([0.5], [t])[0, 0] - A(np.array([0.5 + 1j * t]))[0])
hits = []
for i in cand:
    r = minimize_scalar(lambda t: f(t) ** 2, bounds=(T[i] - 0.01, T[i] + 0.01), method='bounded', options={'xatol': 1e-12})
    hits.append((r.x, f(r.x), d[i]))
zz = [float(mp.im(mp.zetazero(k))) for k in range(1, 30)]
zz = [z for z in zz if z <= 100]
log('zeros of zeta with 0 < gamma <= 100: %d' % len(zz))
log('  t* (G = A)          min |G-A|    nearest gamma_k      |t* - gamma_k|   G(1/2+it*)')
rows = []
for x, v, _ in hits:
    if v > 1e-6:
        log('  local minimum at t=%.4f with |G-A|=%.3e (not a coincidence)' % (x, v)); continue
    k = int(np.argmin([abs(x - z) for z in zz])); gv = G_grid([0.5], [x])[0, 0]
    rows.append([k + 1, x, zz[k], v, gv.real, gv.imag])
    log('  %.10f  %.2e   %.10f   %.1e    %.6f%+.6fi' % (x, v, zz[k], abs(x - zz[k]), gv.real, gv.imag))
log('coincidences found: %d; every zero of zeta up to 100 matched: %s' % (len(rows), sorted(r[0] for r in rows) == list(range(1, len(zz) + 1))))
r4 = lambda v: [float('%.5g' % x) for x in v]
json.dump(dict(t=[float(x) for x in T[::2]], Gr=r4(g.real[::2]), Gi=r4(g.imag[::2]), Ar=r4(a.real[::2]), Ai=r4(a.imag[::2]),
               hits=rows), open('vertical_cut.json', 'w'), separators=(',', ':'))
log('time %.1f s' % (time.time() - t0))
```